Performance comparison of Transformer-ML models with corresponding E2E transformer models

Cls. head : Classification head
E2E : End-to-End model

1. Sports News dataset.

Model Type	Classifier	Accuracy	Precision	Recall	F1 (weighted)	F1(macro)	AUC	Inference time (ms/sample)
IndicBERT (E2E)	Cls. head	0.8847 ± 0.0030	0.8840 ± 0.0029	0.8845 ± 0.0030	0.8838 ± 0.0030	0.8824 ± 0.0033	0.9642+- 0.0020	7.2665 ± 0.0572
IndicBERT Backbone	SVM	0.8846 ± 0.0022	0.8844 ± 0.0022	0.8846 ± 0.0022	0.8844 ± 0.0022	0.8832 ± 0.0024	0.9604 ± 0.0025	6.985 ± 0.117
	LR	0.8841 ± 0.0023	0.8838 ± 0.0023	0.8841 ± 0.0023	0.8839 ± 0.0023	0.8825 ± 0.0026	0.9612 ± 0.0017	6.369 ± 0.004
	KNN	0.8847 ± 0.0012	0.8843 ± 0.0013	0.8847 ± 0.0012	0.8843 ± 0.0013	0.8830 ± 0.0013	0.9344 ± 0.0034	8.491 ± 0.041
	GNB	0.8831 ± 0.0024	0.8852 ± 0.0025	0.8831 ± 0.0024	0.8837 ± 0.0024	0.8824 ± 0.0025	0.9252 ± 0.0019	7.049 ± 0.023
	DT	0.8800 ± 0.0056	0.8798 ± 0.0055	0.8800 ± 0.0056	0.8798 ± 0.0055	0.8786 ± 0.0056	0.9364 ± 0.0020	7.005 ± 0.018
	XGB	0.8840 ± 0.0022	0.8837 ± 0.0021	0.8840 ± 0.0022	0.8838 ± 0.0021	0.8825 ± 0.0020	0.9595 ± 0.0034	7.004 ± 0.009
	Ridge	0.8857 ± 0.0015	0.8856 ± 0.0016	0.8857 ± 0.0015	0.8855 ± 0.0016	0.8844 ± 0.0017	0.9584 ± 0.0005	6.728 ± 0.025
	RF	0.8839 ± 0.0031	0.8837 ± 0.0030	0.8839 ± 0.0031	0.8837 ± 0.0031	0.8822 ± 0.0031	0.9603 ± 0.0021	6.827 ± 0.012
MuRIL (E2E)	Cls. head	0.8726 +- 0.0033	0.8714 +- 0.0030	0.8719 ± 0.0030	0.8709 ± 0.0035	0.8707 ± 0.0037	0.9563 ± 0.0008	6.9677 ± 0.5312
MuRIL Backbone	SVM	0.8724 ± 0.0025	0.8718 ± 0.0025	0.8724 ± 0.0025	0.8720 ± 0.0024	0.8720 ± 0.0023	0.9266 ± 0.0086	6.906 ± 0.013
	LR	0.8729	0.8724	0.8729	0.8725	0.8726	0.9570 ±	6.772 ±

		± 0.0031	± 0.0029	± 0.0031	± 0.0030	± 0.0029	0.0021	0.018
	KNN	0.8717 ± 0.0026	0.8709 ± 0.0025	0.8717 ± 0.0026	0.8710 ± 0.0025	0.8709 ± 0.0026	0.9217 ± 0.0054	6.863 ± 0.016
	GNB	0.8729 ± 0.0033	0.8725 ± 0.0033	0.8729 ± 0.0033	0.8726 ± 0.0033	0.8726 ± 0.0029	0.9150 ± 0.0023	6.800± 0.017
	DT	0.8709 ± 0.0039	0.8705 ± 0.0038	0.8709 ± 0.0039	0.8705 ± 0.0039	0.8705 ± 0.0038	0.9290 ± 0.0036	6.769± 0.018
	XGB	0.8724 ± 0.0037	0.8719 ± 0.0035	0.8724 ± 0.0037	0.8718 ± 0.0037	0.8719 ± 0.0038	0.9431 ± 0.0092	6.770± 0.016
	Ridge	0.8729 ± 0.0022	0.8724 ± 0.0022	0.8729 ± 0.0022	0.8725 ± 0.0021	0.8724 ± 0.0021	0.9441 ± 0.0057	6.773 ± 0.016
	RF	0.8732 ± 0.0055	0.8728 ± 0.0053	0.8732 ± 0.0055	0.8726 ± 0.0054	0.8727 ± 0.0054	0.9482 ± 0.0038	6.876± 0.012
mBERT (E2E)	Cls. head	0.8276 ± 0.0065	0.8290 +- 0.0045	0.8268 +_0.0063	0.8268 +- 0.0068	0.8252 +- 0.0073	0.9329+- 0.0011	7.4235 ± 0.2848
mBERT Backbone	SVM	0.8266 ± 0.0037	0.8274 ± 0.0041	0.8266 ± 0.0037	0.8268 ± 0.0038	0.8246 ± 0.0042	0.9228 ± 0.0008	6.781 ± 0.052
	LR	0.8286 ± 0.0044	0.8293 ± 0.0043	0.8286 ± 0.0044	0.8287 ± 0.0044	0.8266 ± 0.0046	0.9287 ± 0.0034	6.545 ± 0.009
	KNN	0.8292 ± 0.0049	0.8299 ± 0.0049	0.8292 ± 0.0049	0.8294 ± 0.0049	0.8272 ± 0.0052	0.8942 ± 0.0044	7.369 ± 0.887
	GNB	0.8239 ± 0.0043	0.8281 ± 0.0048	0.8239 ± 0.0043	0.8250 ± 0.0044	0.8228 ± 0.0048	0.8844 ± 0.0041	6.559 ± 0.002
	DT	0.8189 ± 0.0039	0.8202 ± 0.0041	0.8189 ± 0.0039	0.8191 ± 0.0038	0.8170 ± 0.0036	0.8982 ± 0.0045	6.559 ± 0.017
	XGB	0.8263 ± 0.0034	0.8273 ± 0.0036	0.8263 ± 0.0034	0.8265 ± 0.0035	0.8244 ± 0.0036	0.9278 ± 0.0041	6.538 ± 0.014
	Ridge	0.8286 ± 0.0048	0.8294 ± 0.0048	0.8286 ± 0.0048	0.8289 ± 0.0048	0.8268 ± 0.0052	0.9298 ± 0.0007	6.558 ± 0.015

	RF	0.8244 ± 0.0042	0.8252 ± 0.0044	0.8244 ± 0.0042	0.8246 ± 0.0042	0.8224 ± 0.0043	0.9220 ± 0.0065	6.627 ± 0.013
DistilBERT (E2E)	Cls. head	0.8016 +_0.0074	0.8021 +_0.0068	0.8012 +_0.0074	0.8004 +_0.0091	0.7978 +- 0.0099	0.9192+- 0.0025	4.0047 ± 0.1913
DistilBERT Backbone	SVM	0.7943 ± 0.0017	0.7945 ± 0.0017	0.7943 ± 0.0017	0.7941 ± 0.0017	0.7915 ± 0.0017	0.9090 ± 0.0025	4.089 ± 0.059
	LR	0.7929 ± 0.0038	0.7932 ± 0.0042	0.7929 ± 0.0038	0.7929 ± 0.0039	0.7909 ± 0.0039	0.9113 ± 0.0049	3.773 ± 0.005
	KNN	0.7969 ± 0.0043	0.7976 ± 0.0052	0.7969 ± 0.0043	0.7966 ± 0.0049	0.7933 ± 0.0049	0.8788 ± 0.0057	4.921 ± 0.872
	GNB	0.7891 ± 0.0053	0.7916 ± 0.0045	0.7891 ± 0.0053	0.7897 ± 0.0052	0.7866 ± 0.0051	0.8478 ± 0.0062	3.802 ± 0.001
	DT	0.7860 ± 0.0033	0.7863 ± 0.0036	0.7860 ± 0.0033	0.7857 ± 0.0033	0.7828 ± 0.0031	0.8671 ± 0.0061	3.776 ± 0.002
	XGB	0.7964 ± 0.0055	0.7966 ± 0.0053	0.7964 ± 0.0055	0.7962 ± 0.0054	0.7936 ± 0.0053	0.9119 ± 0.0022	3.507 ± 0.083
	Ridge	0.8012 ± 0.0054	0.8020 ± 0.0055	0.8012 ± 0.0054	0.8014 ± 0.0055	0.7990 ± 0.0058	0.9118 ± 0.0019	3.777 ± 0.003
	RF	0.7949 ± 0.0042	0.7951 ± 0.0039	0.7949 ± 0.0042	0.7946 ± 0.0040	0.7916 ± 0.0042	0.9056 ± 0.0050	3.845 ± 0.019
XLMR (E2E)	Cls. head	0.8644 +_0.0064	0.8636 +- 0.0055	0.8633 ± 0.0062	0.8621 ± 0.0061	0.8614 ± 0.0066	0.9565 ± 0.0032	6.5287 ± 0.0204
XLMR Backbone	SVM	0.8695 ± 0.0042	0.8690 ± 0.0041	0.8695 ± 0.0042	0.8690 ± 0.0042	0.8687 ± 0.0040	0.9466 ± 0.0021	6.553 ± 0.025
	LR	0.8688 ± 0.0054	0.8682 ± 0.0055	0.8685 ± 0.0061	0.8679 ± 0.0061	0.8674 ± 0.0059	0.9556 ± 0.0024	6.335 ± 0.014
	KNN	0.8688 ± 0.0054	0.8682 ± 0.0055	0.8688 ± 0.0054	0.8682 ± 0.0054	0.8678 ± 0.0051	0.9179 ± 0.0023	7.540 ± 0.928
	GNB	0.8676 ±	0.8689 ±	0.8676 ±	0.8679 ±	0.8675 ±	0.9123 ± 0.0033	6.364 ± 0.014

		0.0054	0.0050	0.0054	0.0053	0.0050		
	DT	0.8663 ± 0.0038	0.8659 ± 0.0037	0.8663 ± 0.0038	0.8659 ± 0.0038	0.8653 ± 0.0036	0.9207 ± 0.0044	6.338 ± 0.013
	XGB	0.8669 ± 0.0045	0.8664 ± 0.0045	0.8669 ± 0.0045	0.8664 ± 0.0045	0.8661 ± 0.0041	0.9423 ± 0.0079	6.344 ± 0.014
	Ridge	0.8696 ± 0.0053	0.8693 ± 0.0053	0.8696 ± 0.0053	0.8693 ± 0.0053	0.8691 ± 0.0052	0.9473 ± 0.0041	6.331 ± 0.006
	RF	0.8677 ± 0.0041	0.8673 ± 0.0041	0.8677 ± 0.0041	0.8672 ± 0.0042	0.8668 ± 0.0040	0.9387 ± 0.0044	6.415 ± 0.036

2. Dravidian CodeMix dataset

Model Type	Classifier	Accuracy	Precision	Recall	F1 (weighted)	F1(macro)	AUC	Inference time (ms/sample)
IndicBERT (E2E)	Cls. head	0.6423 ± 0.0634	0.6518 ± 0.0596	0.6423 ± 0.0632	0.6404 ± 0.0675	0.5620 ± 0.0607	0.8259 ± 0.0444	6.9801 ± 0.0719
IndicBERT Backbone	SVM	0.6605 ± 0.0464	0.6647 ± 0.0334	0.6605 ± 0.0464	0.6494 ± 0.0550	0.5712 ± 0.0483	0.8198 ± 0.0410	9.307 ± 1.725
	LR	0.6195 ± 0.0894	0.6715 ± 0.0195	0.6195 ± 0.0894	0.6293 ± 0.0738	0.5528 ± 0.0630	0.8246 ± 0.0384	6.586 ± 0.031
	KNN	0.6555 ± 0.0486	0.6521 ± 0.0495	0.6555 ± 0.0486	0.6523 ± 0.0489	0.5717 ± 0.0452	0.8048 ± 0.0238	10.339 ± 0.157
	GNB	0.6321 ± 0.0507	0.6530 ± 0.0350	0.6321 ± 0.0507	0.6319 ± 0.0546	0.5593 ± 0.0435	0.7660 ± 0.0395	6.636 ± 0.030
	DT	0.6400 ± 0.0387	0.6335 ± 0.0394	0.6400 ± 0.0387	0.6325 ± 0.0432	0.5493 ± 0.0421	0.7780 ± 0.0356	6.601 ± 0.027
	XGB	0.6688 ± 0.0373	0.6623 ± 0.0359	0.6688 ± 0.0373	0.6605 ± 0.0408	0.5750 ± 0.0380	0.8331 ± 0.0307	6.616 ± 0.039
	Ridge	0.6154 ± 0.0908	0.6760 ± 0.0211	0.6154 ± 0.0908	0.6282 ± 0.0771	0.5551 ± 0.0666	0.8197 ± 0.0360	6.598 ± 0.028
	RF	0.6679 ±	0.6655 ± 0.032	0.6679 ±	0.6615 ±	0.5803 ±	0.8338 ±	6.669 ± 0.027

		0.0333		0.0333	0.0381	0.0362	0.0272	
MuRIL (E2E)	Cls. head	0.6806 ± 0.0063	0.6867 ± 0.0030	0.6807 ± 0.0063	0.6827 ± 0.0042	0.6041 ± 0.0040	0.8428 ± 0.0033	7.1349 ± 0.0286
MuRIL Backbone	SVM	0.6902 ± 0.0022	0.6859 ± 0.0020	0.6902 ± 0.0022	0.6872 ± 0.0018	0.6089 ± 0.0025	0.7980 ± 0.0059	7.201 ± 0.293
	LR	0.6880 ± 0.0047	0.6860 ± 0.0024	0.6880 ± 0.0047	0.6865 ± 0.0031	0.6079 ± 0.0040	0.8367 ± 0.0043	7.156± 0.019
	KNN	0.6882 ± 0.0033	0.6850 ± 0.0017	0.6882 ± 0.0033	0.6860 ± 0.0020	0.6069 ± 0.0048	0.8031 ± 0.0054	8.907± 0.015
	GNB	0.6864 ± 0.0029	0.6862 ± 0.0022	0.6864 ± 0.0029	0.6857 ± 0.0015	0.6078 ± 0.0023	0.7706 ± 0.0042	7.185± 0.019
	DT	0.6815 ± 0.0025	0.6755 ± 0.0031	0.6815 ± 0.0025	0.6769 ± 0.0027	0.5938 ± 0.0034	0.8008 ± 0.0080	7.133± 0.012
	XGB	0.6920 ± 0.0022	0.6857 ± 0.0032	0.6920 ± 0.0022	0.6869 ± 0.0034	0.6061 ± 0.0056	0.8391 ± 0.0045	7.140± 0.017
	Ridge	0.6858 ± 0.0040	0.6868 ± 0.0022	0.6858 ± 0.0040	0.6861 ± 0.0027	0.6075 ± 0.0032	0.8328 ± 0.0031	7.134± 0.018
	RF	0.6906 ± 0.0027	0.6862 ± 0.0023	0.6906 ± 0.0027	0.6874 ± 0.0022	0.6085 ± 0.0023	0.8399 ± 0.0041	7.200± 0.029
mBERT (E2E)	Cls. head	0.6896 ± 0.0033	0.6917 ± 0.0038	0.6898 ± 0.0032	0.6892 ± 0.0020	0.6099 ± 0.0066	0.8561 ± 0.0016	7.1691 ± 0.0112
mBERT Backbone	SVM	0.6930 ± 0.0028	0.6886 ± 0.0038	0.6930 ± 0.0028	0.6900 ± 0.0033	0.6081 ± 0.0039	0.8468 ± 0.0044	7.411 ± 0.157
	LR	0.6900 ± 0.0025	0.6873 ± 0.0021	0.6900 ± 0.0025	0.6882 ± 0.0019	0.6072 ± 0.0059	0.8442 ± 0.0025	6.546 ± 0.006
	KNN	0.6924 ± 0.0045	0.6906 ± 0.0034	0.6924 ± 0.0045	0.6909 ± 0.0033	0.6122 ± 0.0044	0.8122 ± 0.0048	10.285 ± 0.053
	GNB	0.6762 ± 0.0099	0.6799 ± 0.0075	0.6762 ± 0.0099	0.6769 ± 0.0094	0.5996 ± 0.0075	0.7929 ± 0.0126	6.572 ± 0.011
	DT	0.6815 ± 0.0046	0.6777 ± 0.0043	0.6815 ± 0.0046	0.6789 ± 0.0041	0.6012 ± 0.0042	0.8042 ± 0.0038	6.536 ± 0.009
	XGB	0.6933 ±	0.6881 ±	0.6933 ±	0.6897 ±	0.6072 ±	0.8545 ±	6.550 ± 0.003

		0.0048	0.0053	0.0048	0.0050	0.0054	0.0009	
	Ridge	0.6824 ± 0.0026	0.6870 ± 0.0017	0.6824 ± 0.0026	0.6843 ± 0.0020	0.6066 ± 0.0041	0.8456 ± 0.0025	6.537 ± 0.007
	RF	0.6912 ± 0.0036	0.6894 ± 0.0037	0.6912 ± 0.0036	0.6899 ± 0.0034	0.6115 ± 0.0025	0.8499 ± 0.0027	6.593 ± 0.018
DistilBERT (E2E)	Cls. head	0.6641 ± 0.0076	0.6704 ± 0.0041	0.6643 ± 0.0074	0.6657 ± 0.0038	0.5812 ± 0.0055	0.8415 ± 0.0026	3.8538 ± 0.0396
DistilBERT Backbone	SVM	0.6760 ± 0.0042	0.6703 ± 0.0051	0.6760 ± 0.0042	0.6721 ± 0.0049	0.5901 ± 0.0081	0.8307 ± 0.0055	5.230 ± 0.192
	LR	0.6719 ± 0.0031	0.6687 ± 0.0043	0.6719 ± 0.0031	0.6698 ± 0.0036	0.5843 ± 0.0070	0.8334 ± 0.0031	4.063 ± 0.019
	KNN	0.6685 ± 0.0033	0.6657 ± 0.0035	0.6685 ± 0.0033	0.6668 ± 0.0033	0.5850 ± 0.0067	0.8041 ± 0.0041	6.429 ± 1.820
	GNB	0.6492 ± 0.0031	0.6561 ± 0.0040	0.6492 ± 0.0031	0.6497 ± 0.0048	0.5692 ± 0.0091	0.7994 ± 0.0032	4.063 ± 0.019
	DT	0.6448 ± 0.0060	0.6439 ± 0.0057	0.6448 ± 0.0060	0.6440 ± 0.0057	0.5628 ± 0.0042	0.7202 ± 0.0070	4.045 ± 0.032
	XGB	0.6770 ± 0.0047	0.6703 ± 0.0056	0.6770 ± 0.0047	0.6723 ± 0.0054	0.5879 ± 0.0101	0.8386 ± 0.0017	4.066 ± 0.019
	Ridge	0.6625 ± 0.0062	0.6674 ± 0.0071	0.6625 ± 0.0062	0.6647 ± 0.0066	0.5834 ± 0.0110	0.8320 ± 0.0033	4.060 ± 0.011
	RF	0.6737 ± 0.0039	0.6688 ± 0.0059	0.6737 ± 0.0039	0.6704 ± 0.0054	0.5887 ± 0.0095	0.8386 ± 0.0014	4.074 ± 0.020
XLMR (E2E)	Cls. head	0.6944 ± 0.0085	0.6977 ± 0.0032	0.6946 ± 0.0087	0.6948 ± 0.0065	0.6127 ± 0.0076	0.8601 ± 0.0020	7.1611 ± 0.0296
XLMR Backbone	SVM	0.7044 ± 0.0016	0.6982 ± 0.0014	0.7044 ± 0.0016	0.7001 ± 0.0015	0.6168 ± 0.0018	0.8493 ± 0.0045	7.713 ± 0.319
	LR	0.7013 ± 0.0044	0.6973 ± 0.0049	0.7013 ± 0.0044	0.6988 ± 0.0045	0.6160 ± 0.0048	0.8582 ± 0.0010	6.941 ± 0.006
	KNN	0.6996 ± 0.0036	0.6970 ± 0.0028	0.6996 ± 0.0036	0.6977 ± 0.0031	0.6141 ± 0.0047	0.8112 ± 0.0054	10.014 ± 1.473
	GNB	0.6911 ± 0.0041	0.6942 ± 0.0029	0.6911 ± 0.0041	0.6923 ± 0.0034	0.6095 ± 0.0060	0.7911 ± 0.0010	6.978 ± 0.021

	DT	0.6947 ± 0.0024	0.6881 ± 0.0027	0.6947 ± 0.0024	0.6900 ± 0.0025	0.6038 ± 0.0037	0.8128 ± 0.0083	6.943 ± 0.016
	XGB	0.7040 ± 0.0066	0.6965 ± 0.0063	0.7040 ± 0.0066	0.6978 ± 0.0068	0.6129 ± 0.0061	0.8499 ± 0.0082	6.966 ± 0.019
	Ridge	0.6972 ± 0.0067	0.6976 ± 0.0057	0.6972 ± 0.0067	0.6972 ± 0.0061	0.6145 ± 0.0082	0.8513 ± 0.0024	6.945 ± 0.025
	RF	0.7022 ± 0.0049	0.6968 ± 0.0045	0.7022 ± 0.0049	0.6983 ± 0.0045	0.6143 ± 0.0032	0.8491 ± 0.0060	7.004 ± 0.016